NETWORK INFRASTRUCTURE DESIGN FOR SME OFFICE

by

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Introduction

This is the primary Technical and Financial Proposal against request for proposal (RFP) call for proposal - [redacted]. Scope of Work:The proposal details the design, supply, installation, configuration and supporting modern wired (LAN) and wireless (WLAN) network infrastructure for the organization two-floor head office based in Sri Lanka.

[redacted] is a professional services company with around 50 staff. Current Network Environment – Legacy infrastructure with minimal segmentation, no centralized wireless management and a 300 Mbps fiber Internet connection. Network performance, security, and scalability are affected by these limitations.

This proposal responds to all functional and technical requirements defined in the RFP, which includes a fully scalable wired LAN and deployment of a high-performance Wi-Fi 6 wireless network

Scope of the Proposal

The scope of this proposal includes the following components:

- Design and deployment of wired LAN switching infrastructure
- Implementation of Wi-Fi 6 wireless network with optimized coverage
- Deployment of a next-generation firewall with integrated security services
- Structured cabling system using Cat6A standards
- Implementation of network services and centralized management system
- Basic administrator training and documentation

This proposal aims to deliver a secure, reliable, and scalable network infrastructure capable of supporting current business operations while accommodating future growth.

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Section 1 - Project Methodology

The project follows a structured methodology to ensure that the proposed network solution meets the requirements of [redacted] effectively. The approach consists of several key phases, from initial analysis to final validation.

1. **Requirement Analysis**
   In this phase, the client requirements provided in the RFP are carefully analyzed. This includes understanding the number of users, types of applications, performance expectations, and security needs. The office layout and device requirements are also considered to ensure the network design aligns with real-world conditions.

2. **Conceptual Network Design**
   Based on the identified requirements, a high-level network architecture is developed. This includes defining the network topology, selecting appropriate devices such as switches, access points, and firewall, and planning VLAN segmentation. The goal is to create a scalable and efficient design.

3. **Detailed Design and Planning**
   In this phase, the high-level design is refined into a detailed implementation plan. This includes **wired LAN, WLAN, IP addressing schemes, VLAN configurations, security mechanisms, and network management strategies**. Device placement and interconnections are also finalized.

4. **Implementation Planning**
   The necessary hardware, configurations, and deployment steps are identified. This phase ensures that all components required for the network are properly planned before actual deployment.

5. **Testing and Validation**
   The proposed design is evaluated to ensure it meets performance, security, and reliability requirements. Connectivity, coverage, and failover mechanisms are tested to confirm that the network operates as expected.

6. **Deployment and Training**
   After implementation, basic training is provided to the administrator on network monitoring, configuration, and troubleshooting. Documentation and guidelines are also provided for future management.

Section 2 - Requirements & conceptual design

2.1 Understanding requirements

[redacted] runs a two-floor office in Sri Lanka with around 50 employees, and hosts up to 30 guest devices at once, which totals to approximately 80 concurrent users and 120 active endpoints. The single ISP fiber link supplies 300Mbps downlink that must be protected and managed at the edge. Use cases consist of Microsoft Office 365, Web conferencing (Microsoft Teams or Zoom), and light file systems. they all come with a variety of QoS and latency requirements that the network has to support.

This table below shows you how each key RFP requirement was interpreted, its implication on-site and the design decision it influences.

RFP requirement	Interpretation for this site	Design decision
Managed L2/L3 switches with PoE+	APs and IP phones need PoE power on access ports. Two floors need separate access switches with an uplink to aggregation.	1× aggregation switch (MDF, First Floor), 2× access switches (one per floor), all PoE+ capable.
VLAN segmentation: Staff, Admin, Guest, Devices, IoT	50 staff + 30 guests + printers/APs must be isolated from each other. Guest traffic must not reach corporate VLANs.	5 defined VLANs. Inter-VLAN routing only at L3 firewall. Guest VLAN routes to internet only.
Wi-Fi 6 APs, -65 dBm coverage in ≥95% of usable area	Two open-plan floors (25 users each) + meeting rooms, boardroom, café. Walls and partitions cause signal attenuation.	Minimum 4 APs per floor estimated (confirmed by predictive heatmap). Controller-based deployment for roaming.
NGFW ≥1 Gbps stateful throughput, ≥300 Mbps with security services	ISP link is 300 Mbps — firewall must not become a bottleneck even with IPS/URL filtering active.	Select NGFW with ≥300 Mbps threat-protection throughput. Sized for 120 concurrent sessions minimum.
VPN: remote users (SSL/IPsec) + MFA	Staff working from home need secure access to internal resources. Site-to-site capability for future branch offices.	SSL VPN with MFA (TOTP or push) on NGFW. 50-user concurrent VPN license minimum.

QoS for voice/video	Video conferencing (O365/Teams) is latency sensitive. Must be prioritized over bulk file transfers.	DSCP markings: EF for voice (46), AF41 for video (34). QoS policy applied at access switch and WLAN.
Centralized monitoring, config backup, RBAC admin	IT admin (likely 1–2 people) needs to manage all devices from a single dashboard with role-based access.	NMS/SNMP platform selected (e.g. PRTG or vendor cloud dashboard). Config versioning and automated backups daily.
Guest bandwidth: 5–10 Mbps per client, internet-only	Up to 30 concurrent guests. Total guest bandwidth = up to 300 Mbps — must be rate-limited so they don't saturate the ISP link.	Guest SSID rate-limited at 10 Mbps per client. Captive portal with T&C acceptance. L3 isolation — no lateral access.

Requirements interpretation table

According to the office plan we have decided on best places for the devices to be installed.

Floor	Areas	Users	Key network needs
Ground floor	Reception, 2× meeting rooms, open office, café/lobby	25 staff + guest devices	Guest Wi-Fi in café, PoE for APs, meeting room APs for video calls
First floor	Open office, boardroom, printer area, MDF/network closet	25 staff	MDF location for aggregation switch + firewall, boardroom AP for video conferencing, printer VLAN

Site summary

- Full office architectural plan is shown in Appendix A

2.2 - Conceptual design

The first network design keeps things simple, a straightforward hierarchy with an edge firewall, a core switch in the middle, and access switches distributed across two floors. The internet enters through the ISP connection and connects to a Next-Generation Firewall (NGFW). This firewall handles security, NAT, and VPN management. At the center of the network, there is a main Layer 3 switch in the MDF. It manages all VLAN routing and ensures that the different parts of the network can communicate with each other. Each floor has its own access switches to connect users' computers, phones, printers, and all the wireless access points. A wireless controller centrally manages all these access points, allowing users to enjoy seamless Wi-Fi and move around the building without losing signal.

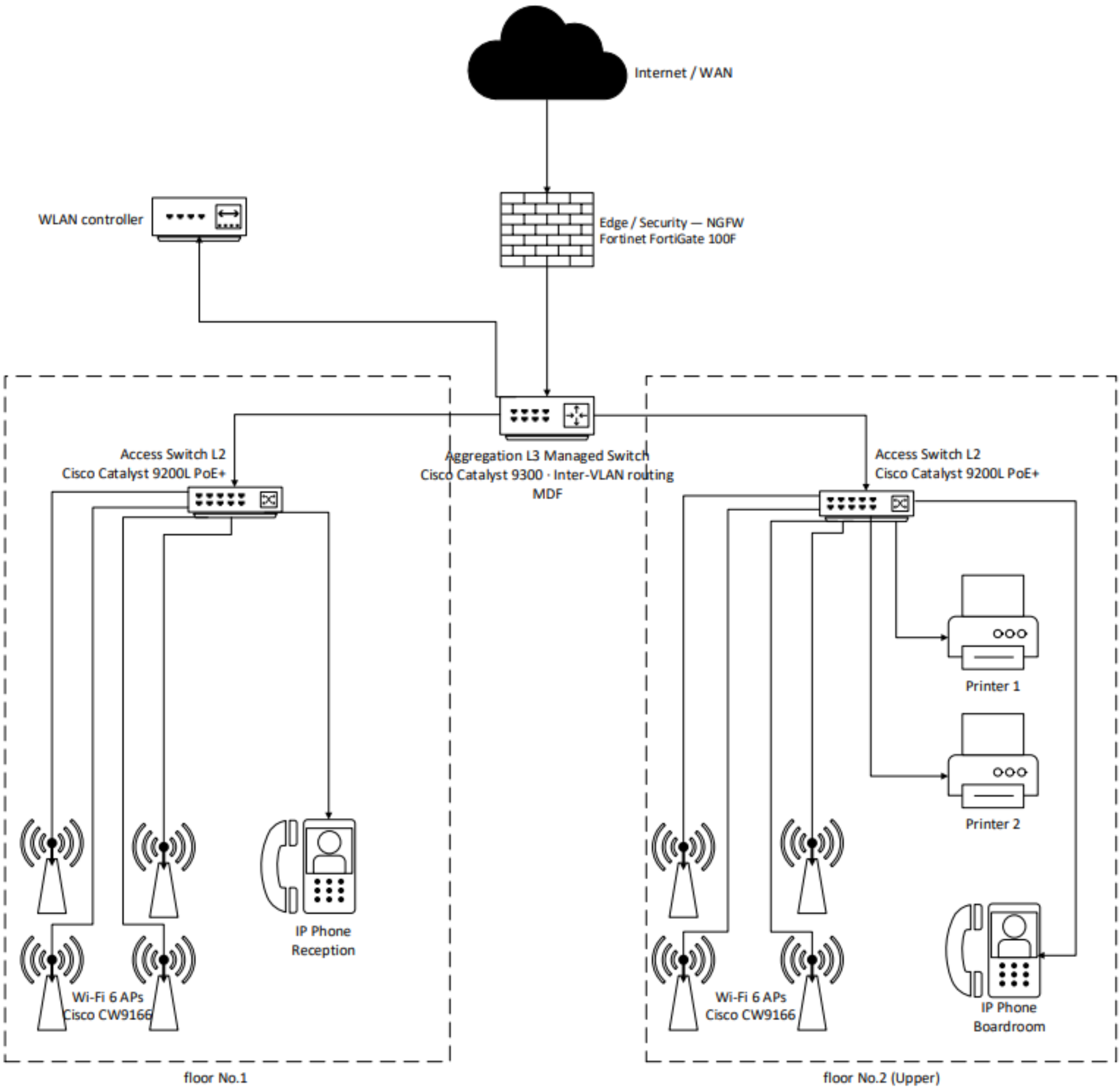

Figure 1: Initial Network Topology

The problem is that this setup lacks real fault tolerance. The main firewall and core switch are both single points of failure. If either one goes down, the entire network can come to a halt instantly.

To solve this, the design shifts to a High Availability (HA) setup. Now, every critical component has a backup—there is device redundancy, extra links, and automatic failover systems. This means that if any piece of hardware or connection fails, the network continues running, users barely notice the issue, and no one is left staring at a “no internet” message. This is how the network remains reliable and operates smoothly, even when hardware fails.

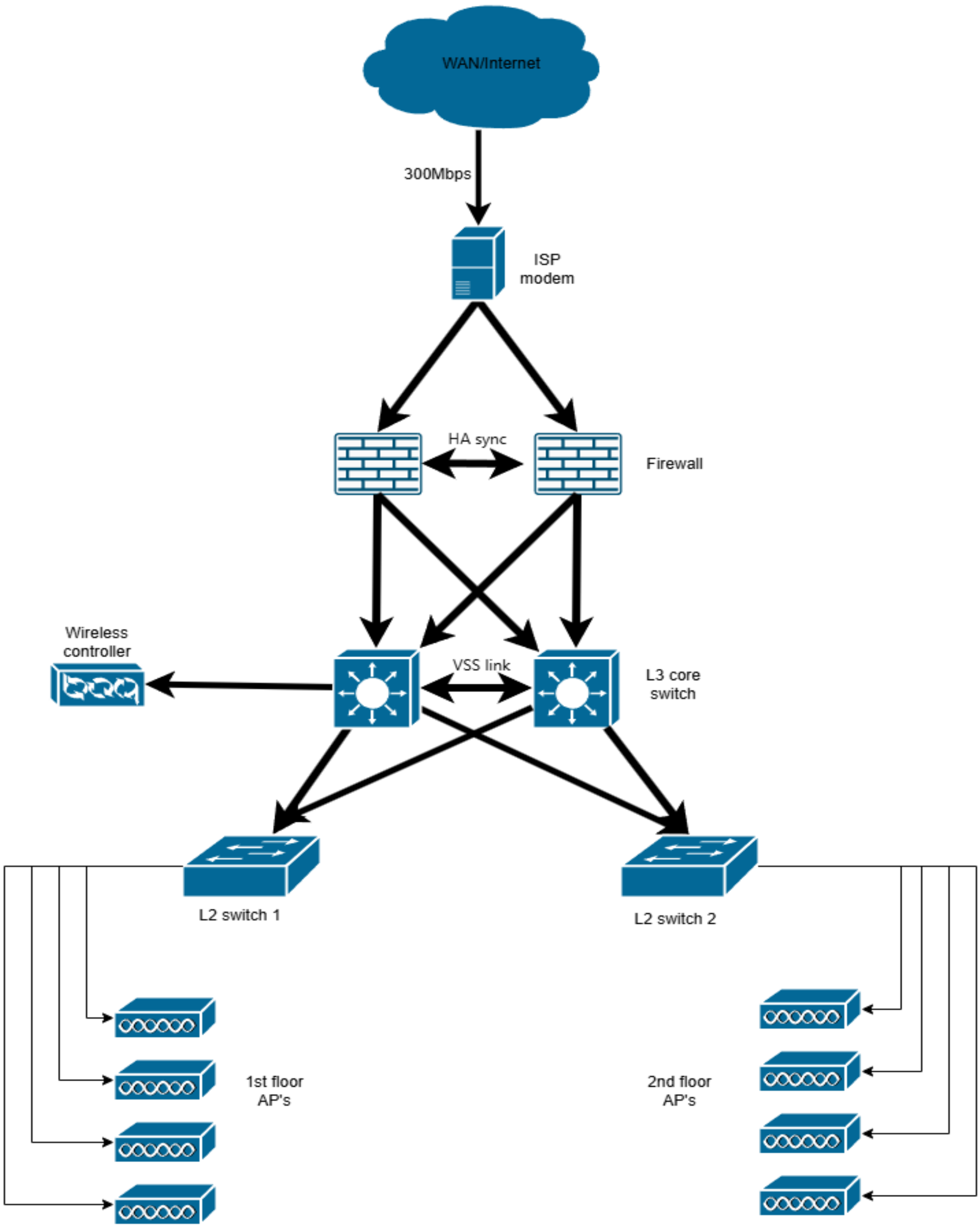

Figure 2: High-Availability Network Architecture

The high-availability network architecture is an optional design choice for the organization. Although it enhances network availability and fault tolerance, it results in higher installation and infrastructure costs.

Architecture layer explanation

Layer	Device(s)	Role in this design
Edge / security	NGFW (e.g., Fortinet FortiGate 100F – HA Pair)	Acts as the redundant gateway to the internet. Perform firewall policy enforcement, stateful inspection, IPS, URL filtering, Network Address Translation (NAT), and VPN (SSL/IPsec) termination. Also enforces Access Control Lists (ACLs) and may act as a DHCP relay for VLANs.
Core / Aggregation	L3 Managed Switches (e.g., Cisco Catalyst 9300 – Dual Core)	Located in the MDF on the first floor. Provides high-speed interconnection between network layers. Performs inter-VLAN routing and acts as the default gateway Supports high-speed (10 Gbps) uplinks to firewalls and aggregated (LACP) links to access switches.
Access –Floor 1	L2 PoE+ Switch (e.g., Cisco Catalyst 9200L)	Provides connectivity to end-user devices, wireless access points, and IP phones on the ground floor. Supports VLAN segmentation, 802.1X port-based authentication, and DHCP snooping for security.
Access –Floor 2	L2 PoE+ Switch (e.g., Cisco Catalyst 9200L)	Same functionality as ground floor access layer, with additional support for administrative devices and printers. Devices are segmented into appropriate VLANs (e.g., Admin VLAN, Printer VLAN).
Wireless	Wi-Fi 6 Access Points + Wireless LAN Controller (e.g., Cisco CW9166 + Controller)	Provides wireless connectivity across both floors. Managed centrally via a wireless LAN controller for configuration, monitoring, and optimization. Supports multiple SSIDs (Corporate and Guest), WPA3 security, and seamless roaming between access points. Traffic is mapped to appropriate VLANs for segmentation and policy enforcement.
Management	Cloud NMS + WLAN Controller	Provides centralized monitoring and management of network devices using SNMP and telemetry. Enables real-time alerts (e.g., WAN failure, AP disconnection), configuration backups, and performance monitoring. Role-Based Access Control (RBAC) ensures secure administrative access.

Section 3 - Wired LAN and Link-Layer Design

3.1 VLAN and Switching Design

The wired LAN for [redacted] is designed as a two-tier switched architecture. A central aggregation switch is in the Main Distribution Frame (MDF) on the first floor, while one access switch serves the ground floor user area and one access switch serves the first-floor user area. This design is appropriate for a two-floor SME office because it is simple, scalable, easy to troubleshoot, and capable of supporting approximately 50 staff members, up to 120 active endpoints, and multiple PoE-powered access points.

A security-first design is used for inter-VLAN communication. Although the MDF switch is Layer 3 capable, in this proposal inter-VLAN routing is centralized at the Next-Generation Firewall (NGFW). This means traffic moving between Staff, Admin, Guest, Network Devices, and IoT segments must pass through firewall policy inspection instead of moving freely between VLANs. As a result, the aggregation switch mainly operates as a high-performance Layer 2 aggregation point, while the NGFW enforces routing policy and security inspection.

VLAN Schema

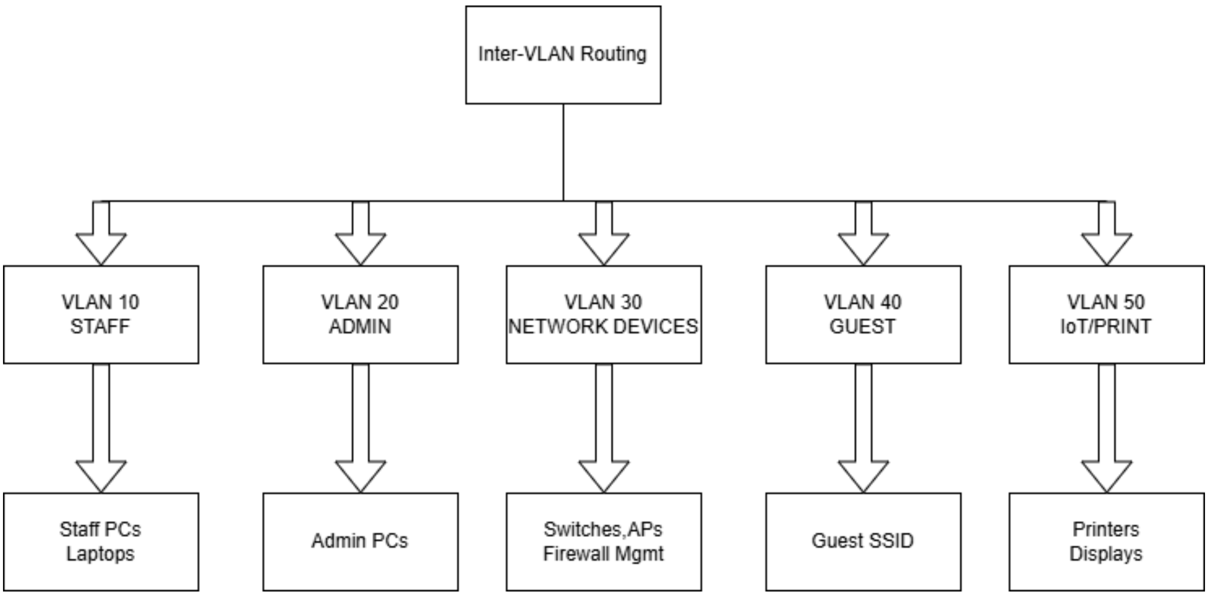

Figure 3.1: Inter VLAN routing

VLAN ID	Name	Subnet	Gateway	DHCP scope	Purpose and members
10	Staff	192.168.10.0/24	192.168.10.1	.10 - .200	All staff PCs, laptops, and corporate wireless clients. Main business-user segment.
20	Admin	192.168.20.0/24	192.168.20.1	.10 - .50	IT admin workstations and privileged management clients. Restricted access.
30	Network Devices	192.168.30.0/24	192.168.30.1	Static/reserved	Switches, access points, and management interfaces. No normal user devices allowed.
40	Guest	192.168.40.0/24	192.168.40.1	.10 - .150	Primarily carries guest SSID traffic from wireless APs. Internet-only access, rate-limited, no internal reachability.
50	IoT / Printers	192.168.50.0/24	192.168.50.1	.10 - .60	Printers, meeting-room displays, and other non-user endpoints with limited internal access.
99	Native / Unused	No client subnet	No SVI	None	Reserved native VLAN on trunk links only. No end devices, no user services, used to reduce VLAN-hopping risk.

Switch selection & justification

Device	Proposed model	Qty	Justification
Aggregation switch (MDF)	Cisco Catalyst 9300-24T	1	Layer-3-capable enterprise switch placed in the MDF. Used here mainly for aggregation and resilient trunking. 10G-capable uplinks support the NGFW and both access switches. No PoE required at this layer.
Access switch - ground floor	Cisco Catalyst 9200L-24P-4X	1	24-port PoE+ access switch for user devices and APs. Supports 802.1X, DHCP snooping, Dynamic ARP Inspection, and 10G-capable uplinks.
Access switch - first floor	Cisco Catalyst 9200L-24P-4X	1	Same model as ground floor for consistency, easier management, and spare-parts simplification. Can be installed in or near the MDF.

Uplink and link aggregation design

Link	Speed	Protocol	Purpose
NGFW -> aggregation switch	10 Gbps (SFP+)	Single trunk	Carries routed VLAN traffic between the internal LAN and firewall. 10G avoids bottlenecks against the 300 Mbps ISP and future internal growth.
Aggregation -> ground floor access switch	2 x 10 Gbps	LACP (IEEE 802.3ad)	Logical aggregated uplink for higher throughput and failover if one link drops.
Aggregation -> first floor access switch	2 x 10 Gbps	LACP (IEEE 802.3ad)	Same design as above. Provides resilience and consistent architecture across both floors.

Trunk and access ports

Port type	VLANs carried	Notes
Trunk ports (switch-to-switch and switch-to-NGFW)	Tagged: 10, 20, 30, 40, 50; native/unused: 99	IEEE 802.1Q tagging is used on trunk links. Native VLAN 99 is reserved and carries no user traffic.
Access ports - staff area	VLAN 10 (untagged)	Each user-facing port belongs to one VLAN. Staff devices authenticate before access is granted.
Access ports - admin area	VLAN 20 (untagged)	Used only for privileged endpoints. Stronger access control applied.
Access ports - printers and IoT	VLAN 50 (untagged)	Non-user devices placed in a restricted segment with limited permitted traffic.
AP uplink ports	Trunk carrying 10, 40, and 50 as needed	APs transport multiple SSIDs over one wired link, so the AP-facing port is configured as a trunk and powered by PoE+.

This switching design meets the assignment requirement for managed switching, VLAN segmentation, PoE support, and resilient uplinks while staying realistic for a two-floor office deployment.

3.5 Link-Layer Security Features

The access layer gets locked down with a mix of security measures: IEEE 802.1X port-based authentication, MAC Authentication Bypass (MAB), DHCP snooping, Dynamic ARP Inspection (DAI), and Rapid Spanning Tree Protocol (RSTP). These tools work together to block some of the most common Layer 2 attacks, whether from a rogue device, someone plugging into a port that's not set up right, or anything else sneaking onto the office network. Why does this matter? Because here, it's not just a list of security buzzwords. The next part breaks down each control—how it works, what threats it stops, and what that looks like in Behpaya's wired LAN setup. It turns abstract concepts into real steps you can follow.

Feature	**Standard / basis**	**Attack prevented**	**Implementation in this design**
802.1X port authentication	IEEE 802.1X	Unauthorized device access through a wall port	Enabled on staff and admin access ports. Devices must authenticate before joining the correct VLAN. Unauthenticated devices are denied normal network access or placed into a restricted state.
MAC Authentication Bypass (MAB)	Vendor-supported fallback method	Allows non-802.1X devices to be identified without disabling edge security	Used for printers and selected IoT endpoints in VLAN 50. Device MAC addresses are validated through policy when full 802.1X supplicant support is unavailable.
DHCP snooping	RFC 2131 behavior enforced at switch level	Rogue DHCP server attacks and false gateway/DNS assignment	Enabled on client VLANs. Only the authorized DHCP path toward the NGFW/server side is trusted. Client-facing ports remain untrusted.
Dynamic ARP Inspection (DAI)	Built on trusted ARP validation using DHCP snooping bindings	ARP spoofing / ARP poisoning / gateway impersonation	Enabled mainly on user VLANs where DHCP snooping bindings are available. Static-only management segments can be protected with trusted-port design or excluded if required.
RSTP and BPDU Guard	IEEE 802.1w / 802.1D concepts	Switching loops, accidental rogue switch insertion	RSTP is enabled to maintain loop-free topology. BPDU Guard is enabled on edge ports so a port can shut down if unexpected bridge protocol traffic is received.
Port security	Switch-level MAC limit policy	MAC flooding and casual misuse of open wall ports	User-facing access ports can be limited to a small number of learned MAC addresses and

			configured to restrict rather than fully disable in case of violation.
Native VLAN hardening	IEEE 802.1Q trunking practice	VLAN hopping through double-tagging scenarios	Native VLAN 99 is unused and carries no end-user traffic. VLAN 1 is not used for client access in this design.

3.6 Traffic Types & Protocols

The wired network carries several types of application traffic, each with different protocol behavior and performance needs. Understanding how application, transport, and network protocols interact helps justify QoS treatment, security policy, and VLAN placement. In a business office, not all traffic should be treated equally: real-time collaboration traffic must be prioritized above large background transfers or best-effort browsing.

Traffic type	**Main protocols**	**Transport**	**Network layer**	**QoS / design note**
Voice (VoIP)	SIP, RTP / SRTP	UDP for media; TCP or UDP may be used for signaling depending on platform	IPv4 unicast	High-priority queue. Low delay, jitter, and packet loss are required.
Video conferencing	HTTPS, WebRTC, RTP / SRTP	Mostly UDP for real-time media, TCP/TLS for control and signaling	IPv4 unicast	Second-highest priority. Important for Teams/Zoom-style meetings.
Microsoft 365 and web access	HTTPS, DNS	TCP with TLS for reliable delivery	IPv4 unicast	Best effort or moderate priority. Sensitive to congestion but less delay-critical than voice.

Internal file transfer	SMB / CIFS, FTP / SFTP if used	TCP	IPv4 unicast	Bulk traffic. Should not starve real-time traffic during congestion.
Network management	SNMP, SSH, Syslog, NTP	UDP for SNMP/Syslog/NTP, TCP for SSH	IPv4, primarily within management VLAN	Low bandwidth but operationally important. Management traffic must remain reachable.
DHCP and DNS	DHCP, DNS	UDP	Broadcast for DHCP discovery, unicast for most DNS	Critical infrastructure services. Needed for all endpoint connectivity.

3.7 Key protocol behaviors explained

Protocol topic	**Explanation in this design**
TCP vs UDP	Voice and video generally use UDP for media because retransmission delay would harm real-time communication. File transfer, web browsing, and many business applications rely on TCP because ordered and reliable delivery matters more than ultra-low latency.
DSCP / DiffServ	QoS markings can be trusted from managed endpoints or set at trusted network boundaries. The access and aggregation layers should preserve or remark approved values, so voice and video receive preferred forwarding treatment.
DNS	All user VLANs rely on DNS to reach cloud services such as Microsoft 365. The firewall can forward queries to secure upstream resolvers and apply content-filtering policy for guest traffic.
NTP	All network devices should synchronize to a common time source. Accurate time is important for logs, troubleshooting, and validating security events.

Although this section uses the OSI model for detailed academic mapping, the same design can also be viewed through the simplified TCP/IP model. In TCP/IP terms, HTTPS, DNS, DHCP, and SNMP operate at the Application layer; TCP and UDP operate at the Transport layer; IPv4, routing, and NAT operate at the Internet layer; and Ethernet, ARP, VLAN tagging, and the physical cabling belong to the Network Access layer.

3.8 Real-World OSI Mapping

The OSI model is useful here because it connects textbook theory to real devices and protocols used in the [redacted] office. The table below maps each OSI layer to the protocols, hardware, and practical behavior found in this design.

The network is explained using the OSI model for academic clarity, while the actual deployed network follows the TCP/IP protocol suite used in real-world IP networking.

OSI layer	**Layer name**	**Protocols / standards in this design**	**Devices operating here**	**What it does in this network**
L7	Application	HTTPS, DNS, DHCP, SNMP, SSH, SIP, SMB	User endpoints, NMS server, NGFW	Supports user applications and management services used by office staff and administrators.
L6	Presentation	TLS, SRTP encryption	Endpoints and NGFW	Handles encryption and format translation where needed, especially for secure web and media traffic.
L5	Session	TLS session management, SIP session control	Endpoints and NGFW	Maintains and coordinates communication sessions such as secure web sessions and call setup/teardown.

L4	Transport	TCP, UDP	NGFW and endpoints	Provides reliable delivery for data traffic and low-latency transport for voice/video traffic.
L3	Network	IPv4, NAT, static routing, default gateways	NGFW primarily; aggregation switch hardware is L3-capable	Handles IP addressing and routing between VLANs through firewall policy.
L2	Data Link	Ethernet, IEEE 802.1Q VLAN tagging, 802.1X, ARP, DHCP snooping, DAI, RSTP	Access switches, aggregation switch, AP Ethernet uplinks	Forwards frames by MAC address, separates VLANs, and applies most access-layer security controls.
L1	Physical	1000BASE-T, 10G SFP+, Cat6 / Cat6A cabling, fiber/copper uplinks	Cables, patch panels, switch ports, transceivers	Carries raw bits across the office cabling system between endpoints, switches, the MDF, and the ISP edge.

3.9 How a typical staff web request moves through the network

Step	Where it happens	OSI focus	What occurs
1	Staff laptop	L7-L4	The user opens a browser and requests an HTTPS page, for example Microsoft 365. The endpoint creates application data and a TCP session toward destination port 443.
2	Staff laptop -> access switch	L2	The laptop sends an untagged Ethernet frame into its access port. Because that switch port belongs to VLAN 10, the switch classifies the traffic into VLAN 10 internally.
3	Access switch uplink	L2	When the frame is forwarded over the trunk uplink toward the aggregation switch, the switch adds an IEEE 802.1Q tag for VLAN 10.
4	Aggregation switch -> NGFW	L2-L3	The aggregation switch forwards the tagged frame across the trunk toward the NGFW. The firewall receives the VLAN traffic and performs inter-VLAN or outbound routing as required.
5	NGFW	L3-L7	The NGFW checks source, destination, NAT policy, and security rules. If permitted, it translates the internal source address and forwards the packet toward the ISP.
6	ISP -> Internet	L1-L3	The packet travels across the WAN to the external service. Response traffic returns to the NGFW public address.
7	NGFW -> user	L3-L2	The firewall matches the returning traffic to the existing session, reverses NAT, and forwards it back on VLAN 10 toward the user.

- The design is technically consistent with a security-first SME office architecture.
- Keeping the OSI mapping is appropriate for the assignment because the marking guide explicitly asks for real-world mapping and protocol understanding.
- A short TCP/IP note is included for clarity, but the OSI model should not be removed from the final submission.

The corrected packet journey now reflects proper switch behavior: access-port traffic is untagged on ingress and tagged only on trunk links

Section 4 - Wireless network design (WLAN)

This section presents the wireless LAN design for [redacted] office design. It covers the **RF planning and AP placement**, **SSID and VLAN mapping**, and **security/authentication design**. The wireless design is aligned with the wired LAN in Section 3 so that both wired and wireless users operate within one consistent VLAN and security architecture.

4.1 RF Planning and AP Placement

A predictive RF planning approach was used for the two-floor office based on the provided floor plan, room layout, user count, and application requirements. The office contains open-plan work areas, reception, café/lobby, two meeting rooms on the ground floor, and an open office, boardroom, printer area, and MDF/network closet on the first floor. The assignment requires wireless coverage of **-65 dBm or better** in at least **95% of usable office areas**, with roaming optimized for voice and video calls.

The proposed design uses **Wi-Fi 6 access points** because they provide better performance in office environments with many connected users. Features such as better spectrum efficiency, improved handling of multiple clients, and stronger support for real-time applications make Wi-Fi 6 suitable for this office. Wi-Fi 6E can also be considered if supported client devices are available.

Based on the office layout, a total of **eight access points** is proposed, with **four APs per floor**. This is also consistent with the wired design, which already reserves PoE capacity for four Wi-Fi 6 APs on each access switch. Four APs per floor are reasonable because the design must cover both open office spaces and enclosed rooms while maintaining good overlap for roaming.

AP Count Justification

Floor	**Coverage areas**	**AP count**	**Placement reason**
Ground Floor	Reception, open office, café/lobby, meeting room 1, meeting room 2	4	One AP for open office, one for reception side, one for café/lobby, one for meeting-room area
First Floor	Open office, boardroom, printer area, MDF/network side	4	Two APs for open office capacity and roaming, one for boardroom, one for printer/MDF side

Floor plan

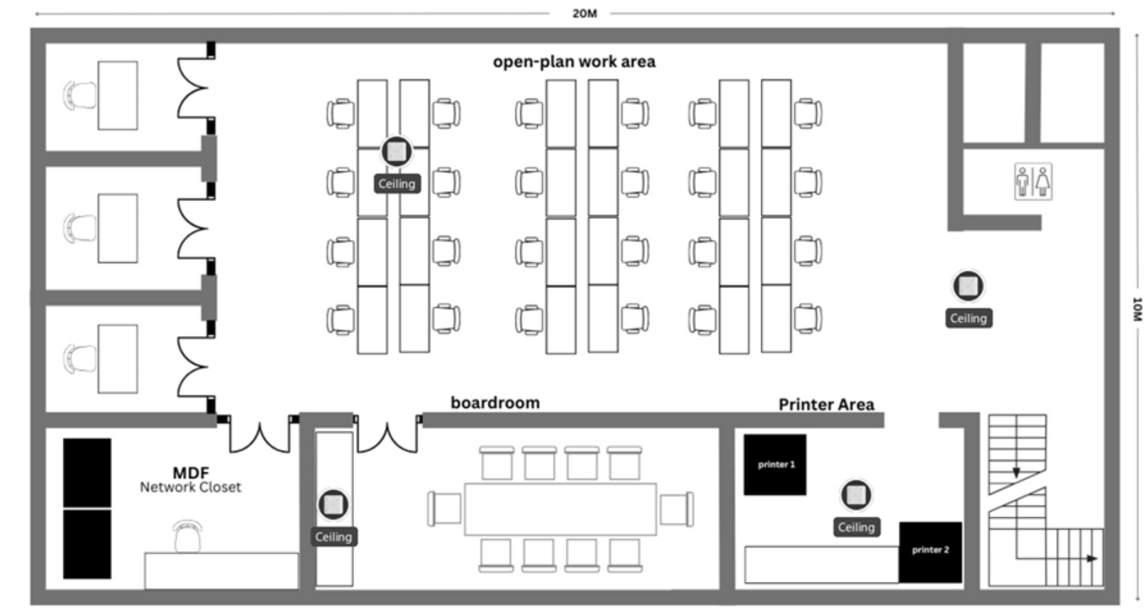

Figure 4.1: First floor

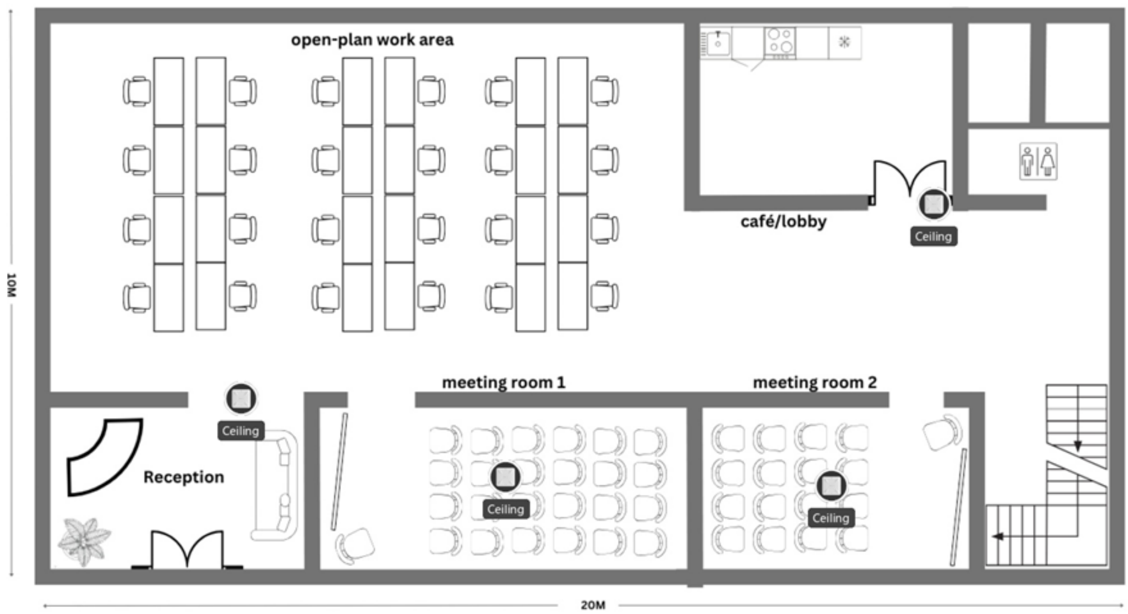

Figure 4.2: Ground floor

4.2 Proposed AP Placement

AP ID	Floor	Proposed location	Main coverage area
AP1	Ground Floor	Centre of open-plan work area	Main staff workspace
AP2	Ground Floor	Near reception side	Reception and left-side entrance area
AP3	Ground Floor	Near café/lobby	Guest/common area
AP4	Ground Floor	Between meeting room 1 and meeting room 2	Meeting rooms and nearby corridor
AP5	First Floor	Left side of open-plan work area	First-floor open office users
AP6	First Floor	Right side of open-plan work area	Extra open-office capacity and roaming overlap
AP7	First Floor	Centre of boardroom	Boardroom and conferencing area
AP8	First Floor	Near printer area / MDF side	Printer area, network side, and edge coverage

All access points will be **ceiling-mounted** and connected to the access switches using **PoE+**. Ceiling mounting improves signal distribution and helps reduce obstruction from furniture or people. The open-plan work areas receive the highest priority because they contain the largest number of users. The meeting rooms and boardroom also receive dedicated AP support because video conferencing and collaboration traffic are more sensitive to delay and packet loss.

4.3 Channel and Power Plan

To reduce interference, the **2.4 GHz band** will use only channels **1, 6, and 11**, while the **5 GHz band** will use different nearby channels across adjacent APs. The 5 GHz band is preferred for employee devices because it gives better performance and less congestion. Medium transmit power is recommended for most APs so that coverage overlaps enough for roaming without creating too much co-channel interference.

AP	2.4 GHz channel	5 GHz channel	Tx power
AP1	1	36	Medium
AP2	6	100	Medium-low
AP3	11	149	Medium-low
AP4	1	44	Medium
AP5	6	36	Medium
AP6	1	100	Medium
AP7	11	149	Medium
AP8	6	44	Low-medium

4.4 SSID and VLAN Mapping

The wireless network uses separate SSIDs for staff, guests, and optional IoT devices. Each SSID is mapped to the correct VLAN defined in Section 2 so that the wireless design remains fully consistent with the wired LAN. This is important because routing, DHCP scopes, firewall policies, and access control are all based on the VLAN structure already defined in the report.

SSID and VLAN Mapping Table

SSID name	VLAN ID	Band	Authentication	Policy	Purpose
BEHPAYA-CORP	10	5 GHz preferred, 2.4 GHz fallback	WPA2/WPA3-Enterprise	Full internal access based on policy	Staff devices
BEHPAYA-GUEST	40	2.4 GHz + 5 GHz	Captive portal	Internet-only, rate-limited, L3 isolation	Visitors and guest devices
BEHPAYA-	50	2.4 GHz	WPA2-PSK	Restricted	Printers and

IOT		mainly		access to approved services only	smart devices
AP / WLAN Management	30	N/A	Internal management only	Device management traffic only	APs and controller

VLAN 10 is used for staff and corporate wireless users,

VLAN 40 is reserved for guest traffic,

VLAN 50 is used for printers and IoT devices,

VLAN 30 is reserved for network devices such as access points and management interfaces.

VLAN 99 remains unused and must not be assigned to any wireless SSID.

The Corporate SSID is mapped to VLAN 10 so that staff users receive the same level of access whether they connect through wired ports or Wi-Fi. The Guest SSID is mapped to VLAN 40 so that guest traffic remains isolated and is sent only toward the firewall for internet access. The optional IoT SSID is mapped to VLAN 50 to separate printers and other non-user devices from staff systems. Management traffic for the APs and wireless controller is kept on VLAN 30 to protect the management plane from normal user traffic.

The AP uplink ports on the access switches should be configured as **802.1Q trunk ports** carrying **VLANs 10, 30, 40, and 50**. This allows one AP to carry multiple SSIDs while remaining managed separately. This is the correct wireless interpretation of the VLAN scheme used in Section 2.

4.5 Security and Authentication Design

The wireless security design follows a zero-trust approach. No device should be allowed normal access to the corporate network unless it is authenticated and placed in the correct VLAN. This supports the assignment requirement for **WPA2/WPA3-Enterprise**, **captive portal guest access**, and proper separation between staff, guest, and IoT devices.

Corporate SSID Security

BEHPAYA-CORP’s Wi-Fi network uses WPA2/WPA3-Enterprise security, combining 802.1X authentication with a RADIUS server. Here’s how it works: your device is the supplicant, the access point handles authentication, and the RADIUS server checks your credentials against the company’s directory. Once you’re verified, you land on VLAN 10 and pick up an IP address from the staff subnet. This setup is way safer than a single shared password. Each user has unique credentials, allowing administrators to quickly revoke or manage access for specific individuals when needed.

The authentication process is straightforward. First, your device connects to the Corporate SSID. At this point, the access point only lets through authentication traffic. Your device sends your credentials using EAP, the AP passes that along to the RADIUS server, and if everything checks out, the server gives

Guest SSID Security

The BEHPAYA-GUEST network uses a captive portal. When guests connect, they get IP addresses from VLAN 40 and are pushed to the portal before they can use the internet. Guests can't reach any internal company resources—just the internet—and the firewall enforces these limitations. This setup fits both the assignment's requirements and the VLAN's purpose outlined earlier.

Each guest's traffic also needs to be capped at 10 Mbps and guarded with Layer 3 client isolation. That way, no single guest can hog bandwidth, and guests can't talk to each other or get into the company's internal VLANs. With the office supporting up to 30 guests at once, these settings make sure the 300 Mbps internet stays reliable for staff.

IoT SSID Security

The optional **BEHPAYA-IOT** SSID is mapped to **VLAN 50** and is intended for printers, meeting-room displays, or other smart devices. These devices often do not support enterprise authentication, so a controlled **WPA2-PSK** approach is acceptable for this VLAN. However, the IoT network must remain isolated from staff and admin systems and should be allowed to reach only the specific internal services it needs, such as print services. This reduces the security risk if a printer or display device is compromised.

Controller Role and Roaming Support

The wireless controller provides centralized configuration, firmware updates, monitoring, RF optimization, and SSID-to-VLAN enforcement. It also helps improve roaming by coordinating channel and power settings across all APs. Roaming assistance features can be enabled where supported so that users moving between APs during voice or video calls experience minimal interruption. This supports the assignment requirement for controller-based management and roaming optimization for voice/video traffic.

- Predictive heatmaps and KPI test tables are shown in Appendix B section.

Section 5 - Security implementation

5.1 Threat identification

The proposed network is exposed to both internal and external threats. These threats are identified based on different network zones such as Staff LAN, Wireless LAN, Guest network, and Internet edge.

Threat	**Zone**	**Likelihood**	**Impact**	**Description**
Malware via email/web	Staff VLAN	High	High	Users may download malicious files leading to data loss or system compromise
Rogue Access Point	WLAN	Medium	High	Unauthorized AP can bypass security and allow attackers into the network
Guest access to internal network	Guest VLAN	Medium	High	Guest users may try to access staff or admin resources
ARP Spoofing	Wired LAN	Medium	High	Attacker can intercept traffic within the LAN
VPN brute-force attack	Edge/VPN	High	High	Attackers attempt login using multiple password attempts
Unauthorized device connection	Wired LAN	Low	High	Unknown device plugged into network ports

5.2 Security Implementation

Security controls are implemented using a layered approach to protect the network from the identified threats.

- **Firewall Rules (NGFW)**
  A next-generation firewall is used to control traffic between VLANs and the internet. By default, all unnecessary traffic is blocked, and only required services are allowed. This prevents unauthorized access and protects internal networks from external attacks.

- **VLAN Segmentation**
  The network is divided into separate VLANs such as Staff, Admin, Guest, and IoT. This limits communication between different groups and prevents threats like malware from spreading across the entire network.

- **WLAN Security (WPA2/WPA3)**
  Wireless networks are secured using WPA2/WPA3 encryption. This ensures that only authorized users can connect and protects against unauthorized access and rogue devices.

- **VPN with Multi-Factor Authentication (MFA)**
  Remote users connect to the network using a secure VPN. MFA is implemented by requiring both a password and a one-time code, reducing the risk of brute-force attacks.

- **Port Security / 802.1X Authentication**
  Network ports are configured to allow only authorized devices. Unauthorized devices connected to switch ports will be blocked, preventing physical access threats.

- **DHCP Snooping and ARP Protection**
  These features help prevent attacks such as ARP spoofing by ensuring that only valid devices can send network information within the LAN.

- **URL Filtering**
  Access to malicious or unsafe websites is blocked to reduce the risk of malware infections through web browsing.

- **Logging and Monitoring**
  Network devices generate logs that are monitored to detect suspicious activities and support troubleshooting.

Section 6 - Network Management System & detailed design

6.1 NMS features & justification

The proposed NMS for [redacted] IT is a combination of the vendor cloud management platform (for WLAN and switch management) and PRTG Network Monitor (for cross-vendor SNMP monitoring, alerting, and syslog aggregation). This hybrid approach avoids vendor lock-in while providing a single-pane-of-glass view of all network devices - switches, NGFW, APs, and servers, from one dashboard.

NMS feature table

Feature	Protocol / method	Justification & operational benefit
Device monitoring & availability	SNMP v3, ICMP ping	Real-time device status monitoring and failure detection
Alerting	SNMP Traps, Email/SMS	Immediate notification for network issues
Interface & traffic monitoring	SNMP MIB-II, NetFlow / sFlow	Track bandwidth usage and detect abnormal traffic
Syslog aggregation	Syslog (UDP 514 / TCP 601)	Provides single searchable log store for security incidents, configuration changes, and fault diagnosis. Critical for post-incident forensics — correlating a security event across switch, NGFW, and RADIUS logs.
Configuration backup	SSH/SCP, TFTP, vendor API	Daily automated backup of all switch, NGFW, and AP controller configurations. Version comparison allows admins to see exactly what changed between backups — essential for change control and disaster recovery. If a device fails, config can be restored in minutes.
NTP synchronization	NTP (UDP 123)	All devices sync to a single internal NTP server (NGFW), which syncs to pool.ntp.org. Consistent timestamps across all devices are essential for log correlation clock drift makes it impossible to reconstruct the sequence of events during a security incident.
RBAC admin access	SSH (CLI), HTTPS (GUI), RADIUS/TACACS+	Two admin roles defined: Network Admin and Helpdesk. Admin sessions authenticated via AD credentials through TACACS+ or RADIUS. All CLI sessions logged — who logged in, when, and what commands were run.
Rogue AP	WLAN controller RF scanning	WLAN controller continuously scans for

detection		unauthorized APs broadcasting in the office RF space. Any SSID not in the approved list generates an alert.
Disaster recovery procedure	Config backup + documented runbook	Documented step-by-step recovery procedure: (1) replace failed hardware, (2) restore config from NMS backup via TFTP/SCP, (3) verify VLAN and routing operation, (4) test connectivity RTO (Recovery Time Objective) target: <4 hours for complete switch failure.

Given that [redacted] IT may need a small IT team (estimated 1–2 administrators), the NMS must minimize manual effort through automation: automated alerts, scheduled config backups, and threshold-based notifications that alert admins before issues affect users.

6.2 Static IP assignment register - network devices

Device	Hostname	IP address	VLAN	Location
Next-gen firewall	BHY-FW-01	192.168.30.2	VLAN 30	MDF rack, 1st floor
Aggregation switch	BHY-SW-AGG-01	192.168.30.10	VLAN 30	MDF rack, 1st floor
Access switch — GF	BHY-SW-ACC-GF	192.168.30.11	VLAN 30	IDF cabinet, ground floor
Access switch — FF	BHY-SW-ACC-FF	192.168.30.12	VLAN 30	MDF rack, 1st floor
AP — GF open office	BHY-AP-GF-01	192.168.30.20	VLAN 30	GF open office, ceiling mount
AP — GF reception	BHY-AP-GF-02	192.168.30.21	VLAN 30	GF reception, ceiling mount
AP — GF café	BHY-AP-GF-03	192.168.30.22	VLAN 30	GF café/lobby, ceiling mount
AP — GF meeting rooms	BHY-AP-GF-04	192.168.30.23	VLAN 30	GF meeting room cluster
AP — FF open office L	BHY-AP-FF-05	192.168.30.24	VLAN 30	FF open office left, ceiling mount
AP — FF boardroom	BHY-AP-FF-06	192.168.30.25	VLAN 30	FF boardroom, ceiling mount
AP — FF open office R	BHY-AP-FF-07	192.168.30.26	VLAN 30	FF open office right, ceiling mount
AP — FF printer area	BHY-AP-FF-08	192.168.30.27	VLAN 30	FF printer area, ceiling mount
NMS server	BHY-NMS-01	192.168.20.10	VLAN 20	MDF rack / VM on

				admin server
RADIUS server	BHY-RADIUS-01	192.168.20.12	VLAN 20	MDF rack / VM on admin server

6.3 DNS & DHCP configuration summary

VLAN	DHCP server	DNS (primary)	DNS (secondary)	Lease time
VLAN 10 Staff	NGFW — scope 192.168.10.50–.200	192.168.10.1 (NGFW — internal resolver)	8.8.8.8	24 hours
VLAN 20 Admin	NGFW — scope 192.168.20.20–.50	192.168.20.1 (NGFW)	8.8.8.8	24 hours
VLAN 30 Devices	Static only — no DHCP	192.168.30.1 (NGFW)	—	N/A
VLAN 40 Guest	NGFW — scope 192.168.40.10–.150	192.168.40.1 (NGFW — content filtered)	—	4 hours
VLAN 50 IoT	NGFW — scope 192.168.50.20–.60	192.168.50.1 (NGFW)	—	8 hours

6.4 Structured cabling summary

Cable type	Standard	Application in this design
Cat6A UTP horizontal runs	TIA-568-C.2, ISO/IEC 11801	All runs from patch panel to wall outlet and from patch panel to APs. Supports 10GBASE-T up to 100m — future-proofs against 10G access layer upgrades. All runs certified with Fluke DSX cable tester. Pass criteria: insertion loss, NEXT, return loss per TIA-568-C.2.
OM3 multimode fiber	ISO/IEC 11801	Inter-floor backbone between GF IDF and FF MDF if copper distance exceeds 90m or EMI is a concern. Supports 10G over 300m. SFP+ transceivers on aggregation switch.

Patch cables	Cat6A (copper), LC-LC (fiber)	All patch panel to switch connections use factory-terminated Cat6A or fiber patch cables. No field-terminated patch cables — reduces installation errors.
Labelling convention	TIA-606-B	Format: [Floor]-[Room]-[Port number]. Example: GF-OPEN-P01 = Ground Floor, Open Office, Port 01. Labels applied at both ends of every cable run and on patch panel ports. Full cable schedule maintained in as-built documentation.

Section 7 - Network Infrastructure Budget Report

- Exchange Rate: 1 USD = 300 LKR
- Prices are estimates based on enterprise reseller averages
- Includes hardware and basic licensing

Core & Security Infrastructure

Device	Model	Qty	Unit USD	Unit LKR	Total USD	Total LKR	Warranty
NGFW (HA Pair)	Fortinet FortiGate 100F	1	4500	1,350,000	9000	2,700,000	1 Year (extendable)
Aggregation Switch	Cisco Catalyst 9300-24T	1	4000	1,200,000	4000	1,200,000	Limited Lifetime

Subtotal: USD 13,000 | LKR 3,900,000

Access Layer Switching

Device	Model	Qty	Unit USD	Unit LKR	Total USD	Total LKR	Warranty
Access Switch	Cisco Catalyst 9200L-24P-4X	2	3000	900,000	6000	1,800,000	Limited Lifetime

Subtotal: USD 6,000 | LKR 1,800,000

Wireless Infrastructure

Device	Model	Qty	Unit USD	Unit LKR	Total USD	Total LKR	Warranty
Wi-Fi 6 Access Point	Ruijie RG-RAP2260	8	250	75,000	2000	600,000	3 Years
Local WLAN Controller	Ruijie RG-WS6008	1	1200	360,000	1200	360,000	1–3 Years

Subtotal: USD 3,200 | LKR 960,000

Network Management & Servers

Device	Model	Qty	Unit USD	Unit LKR	Total USD	Total LKR	Warranty
NMS Server	PRTG / VM	1	1000	300,000	1000	300,000	1 Year
RADIUS Server	Authentication	1	800	240,000	800	240,000	1 Year

Subtotal: USD 1,800 | LKR 540,000

Cabling & Physical Infrastructure

Item	Description	USD	LKR	Warranty
Cat6A Cabling	Full office structured wiring	2500	750,000	10–15 Years
Network Rack	MDF rack	1200	360,000	1 Year
Patch Panels	Termination panels	600	180,000	1 Year
SFP+ Modules	10G uplinks	1000	300,000	1 Year
Patch Cords	Accessories	500	150,000	6 Months

Subtotal: USD 5,800 | LKR 1,740,000

Grand Total

USD: $29,800
LKR: 8,940,000

Budget conclusion

The cost estimates are based on current market pricing from recognized vendors and provide a dependable approximation. Minor variations may occur due to supplier differences, currency changes, and import duties.

Section 8 - Resource Allocation

Resource allocation defines how tasks and responsibilities are distributed among the project team to ensure efficient completion of the network design.

The project is divided into key technical areas, and each team member is assigned a specific responsibility based on the project requirements. This structured allocation ensures that all components of the network are designed with proper focus while maintaining coordination between different sections.

Task	**Assigned Member**	**Responsibility**
Wired Network Design (LAN)	Nisadi G.D.R. - IT24102489	Design of switching architecture, VLAN segmentation, and redundancy mechanisms
Wireless Network Design (WLAN)	Rathnayaka R.M.S.S. - IT24100416	Access point placement, SSID configuration, and wireless coverage planning
Security & NMS Design	Premathilaka H.P.S.M - IT24102483	Threat identification and implementation of security controls, network monitoring system, and access control mechanisms
Network Design Integration & Coordination	Chandrasiri P. G. S. K. - IT24100454	Overall network design development, and ensuring consistency across all sections

Each member is responsible for their assigned section while regularly coordinating with the team to ensure consistency across the overall design. This approach improves efficiency, reduces overlapping, and ensures that all technical requirements are properly addressed.

9. Communication Plan

Effective communication is essential for successful coordination among team members and ensures that the project progresses smoothly without misunderstandings or delays.

The team uses a combination of communication methods to share information and track progress:

- **Regular Team Meetings**
  The team conducts regular discussions to review progress, clarify requirements, and resolve any issues related to the design.
- **Online Communication Tools**
  Platforms such as WhatsApp and Google Meet are used for quick communication, file sharing, and virtual meetings.
- **Task Updates and Coordination**
  Each member provides updates on their assigned tasks, ensuring that all parts of the network design remain consistent and aligned.
- **Issue Resolution**
  Any technical or design-related issues are discussed within the team, and solutions are agreed upon collaboratively.
- **Documentation Sharing**
  All design documents are shared among team members to maintain transparency and ensure that everyone has access to the latest version.

This communication approach helps maintain coordination, improves teamwork, and ensures the successful completion of the project within the given timeframe.

10. Risk Management

Risk management is essential to identify potential issues that may affect the network design and propose mitigation strategies to minimize their impact.

The following risks have been identified along with their possible impact and corresponding mitigation measures:

Risk	**Impact**	**Mitigation**
Network device failure	Network downtime and service interruption	Use reliable devices and maintain configuration backups for quick recovery
ISP connection failures	Network downtime and service interruption	Use a secondary backup internet connection (failover link) and configure automatic failover on the firewall
Security breach	Unauthorized access or data loss	Implement firewall policies, VLAN segmentation, and secure authentication methods
Power failure	Temporary loss of network services	Use Uninterruptible Power Supply (UPS) for critical devices such as firewall and core switch
Misconfiguration	Network connectivity issues or service disruption	Perform proper testing and validation before deployment
Unauthorized device access	Security compromise through physical access	Use port security and authentication mechanisms such as 802.1X
High network traffic / congestion	Reduced network performance	Implement traffic monitoring and bandwidth management policies

By identifying these risks in advance, the proposed network design ensures better reliability, security, and operational stability.

Hand Over Process

Backups for Configurations

All network devices' (switches, firewall, wireless controller, and access points) configurations have been made reliable through secure backups. The backup files are stored using encryption in local as well as remote locations. There are automated backups done on a daily and weekly basis. Version control has been provided to help recover from any disasters that may occur.

Escrow of Admin Credentials and Hand Over

Administrative credentials are securely handed over to two system administrators who are appointed by the clients. These credentials are saved in an encrypted password vault and given to authorized individuals only. Role-Based Access Control (RBAC) helps give different permission levels as per the requirement. Default passwords are reset while Multi-Factor Authentication (MFA) is put in place during the process of handing over.

Documentation Provided

The customer is supplied with comprehensive documentation that covers network maps, IP and VLAN addresses, device list (with serial numbers and warranty details), and backup/recovery procedures. Proper administrator access instructions are also provided for future system management.

Training Session Administrator Guidance

A training session lasting 2 to 4 hours is arranged for the two administrators. In this session, the whole system is thoroughly described, covering network design, device management, monitoring, troubleshooting, and backup/restore operations.

Moreover, an administrator quick start guide is also provided, which consists of instructions on how to use the system daily, from logging in, monitoring, managing devices, taking backups, and troubleshooting common problems.

Acceptance of the Solution

The system's ability to function properly, along with administrator access and documentation, is confirmed by the client. These problems must be sorted out before accepting the solution. Final acceptance means that the project was completed successfully.

Conclusion

This proposal presents a comprehensive and scalable network solution for [redacted] Information Technology, addressing both current operational needs and future growth. The design integrates a secure wired LAN, high-performance Wi-Fi 6 WLAN, and advanced security mechanisms including VLAN segmentation, NGFW protection, and VPN access. Centralized network management, configuration backups, and structured monitoring ensure reliability and efficient administration. The inclusion of redundancy and risk mitigation strategies improves availability and resilience. Additionally, proper documentation, administrator training, and a clear handover process ensure smooth long-term operation. Overall, the proposed solution delivers a secure, high-performing, and manageable network infrastructure aligned with industry best practices.

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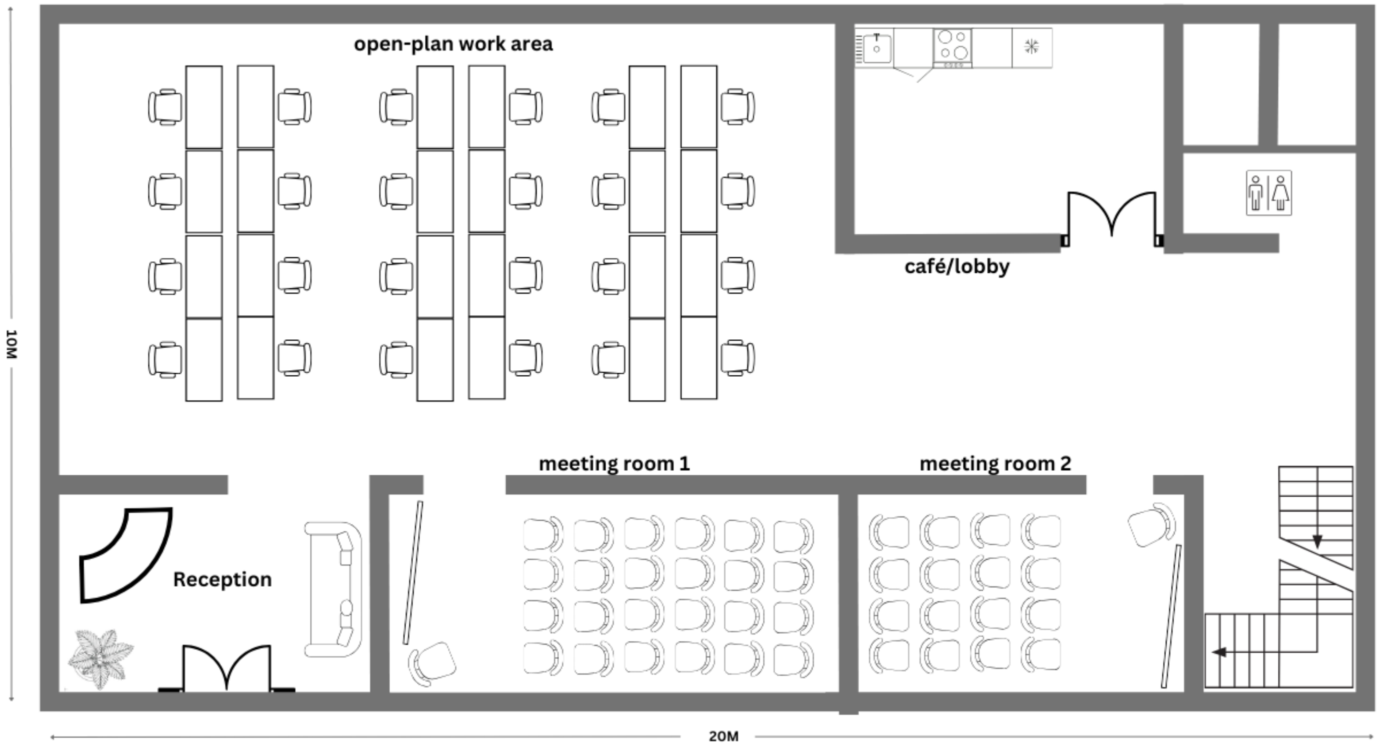
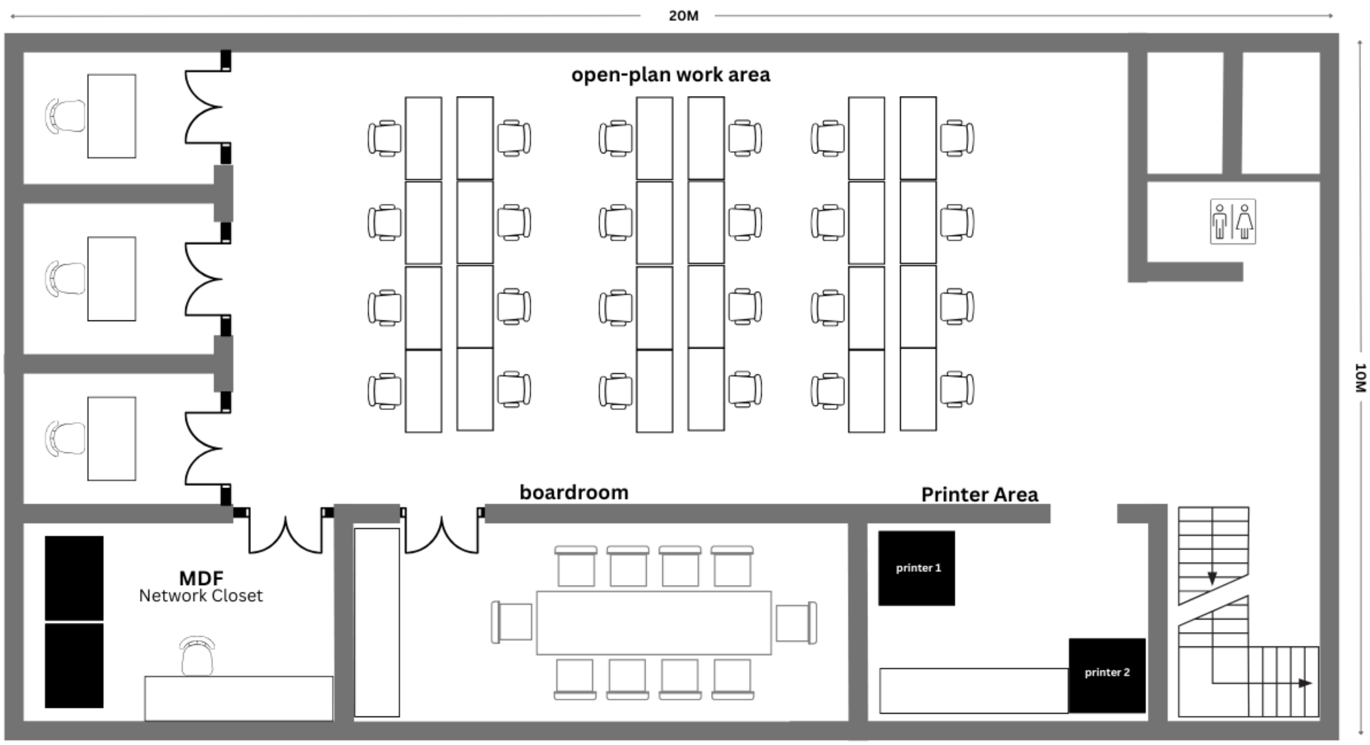
Appendix A – Site Summary & Floor Areas





Appendix B – Device & Capacity Assumptions

User and Device Assumptions

Category	Assumption	Value	Design Impact
Staff users	Total employees	50	Determines VLAN sizing, IP addressing, and switch capacity
Guest users	Concurrent guest devices	Up to 30	equires separate guest VLAN, bandwidth control, and isolation
Total concurrent users	Staff + guests	~80	Impacts wireless capacity and internet usage
Total endpoints	All connected devices	Up to 120	Used for switch port density and DHCP scope design

Application & Traffic Assumptions

Application type	Typical Bandwidth per User	Characteristics	Design Impact
Office 365 / Web	1–2 Mbps	Moderate, bursty	Best-effort traffic
Video conferencing (Teams/Zoom)	2–4 Mbps	Latency-sensitive	Requires QoS prioritization
Web browsing	~1 Mbps	Variable	Standard traffic handling
File services	1–5 Mbps	Bulk transfer	Lower priority than real-time traffic

Firewall & Session Capacity Assumptions

Parameter	Assumption	Value	Design Impact
Sessions per user	Average	50-100	Web and cloud applications
Total concurrent sessions	Estimated	~6000	Firewall sizing
Recommended capacity	With headroom	≥10,000 sessions	Ensures scalability

Growth & Scalability Assumptions

Parameter	Assumption	Design Impact
User growth	10–20% over 2–3 years	Ensures scalability
Future staff	Up to 70 users	VLAN and IP planning
Future endpoints	150+ devices	Switch and wireless capacity

Appendix C – Coverage & Performance KPIs

Figure C.1: 1st floor Heatmap (2.4 GHz)

Figure C.2: 1st floor Heatmap (5 GHz)

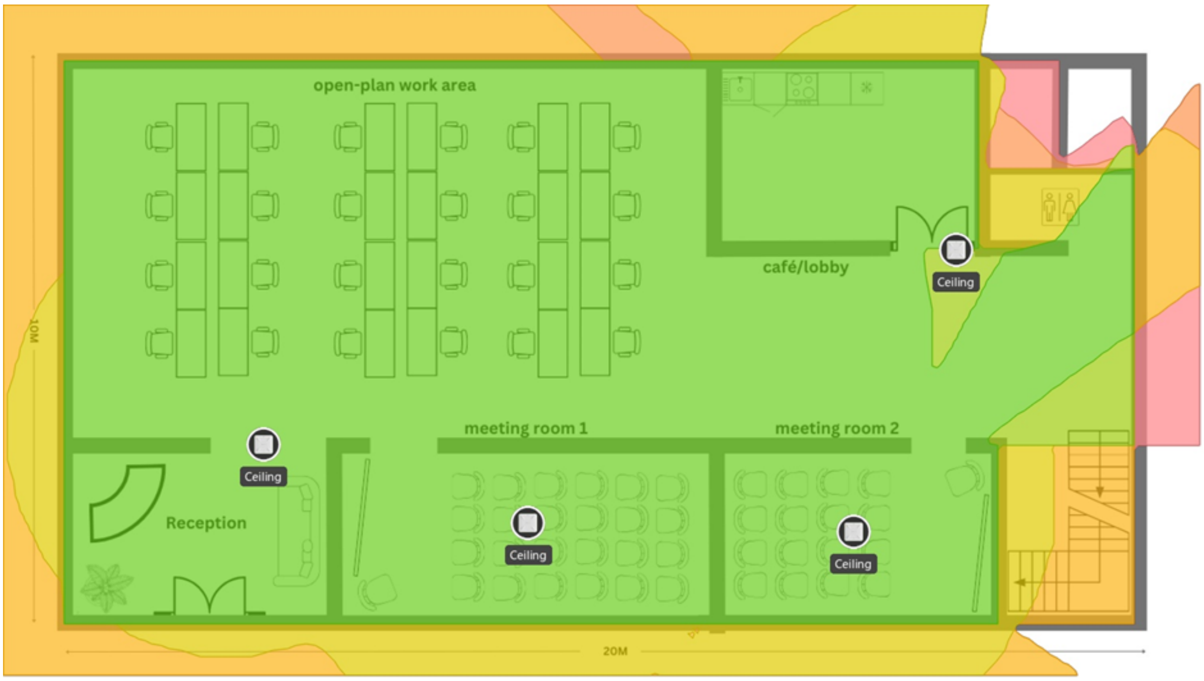


Figure C.3: Ground floor heatmap (2.4 GHz)

Figure C.4: Ground floor Heatmap (5 GHz)